

FINAL SOC INVESTIGATION REPORT

Ryan Adams / FRONTDESK-PC1 - Splunk Capstone

Reporting basis: This report is limited to what the available Splunk telemetry supports. Where causation, attribution, or impact cannot be confirmed, the limitation is stated explicitly.

Findings

Item	Finding
Affected user	Ryan.Adams - unauthorized use of credentials is strongly suspected.
Victim host	FRONTDESK-PC1.KCD.local
Victim IP	172.16.0.110
Authentication source	DESKTOP-924H12 / 172.16.0.184. Windows Security recorded repeated failed and successful NTLM Type 3 authentication activity from this workstation/IP. Multiple usernames were involved, consistent with credential guessing and possible password spraying. Who controlled the source workstation is not established.
Suspicious file	C:\Users\Ryan.Adams\Music\python.exe
File creation	2025-10-15 12:57:00 UTC - chrome.exe created the suspicious file.
Execution	2025-10-15 13:00:33 UTC - python.exe executed from the Music folder.
Suspected external C2	157.245.46.190:8888 - Sysmon attributed the connection to python.exe; Zeek independently confirmed an established TLSv1.3 session to the same IP and port.
Internal activity	python.exe resolved ADDC01.KCD.local to 172.16.0.7 and connected to TCP 135 and 49669, consistent with RPC-related communication. Successful lateral movement was not confirmed.
Persistence	Scheduled task PythonUpdate created at 13:04:59 UTC to run python.exe at system startup as SYSTEM.
Defender state	13:05:53 UTC - Windows Security Center reported SECURITY_PRODUCT_STATE_SNOOZED. The responsible actor/process is not established.
Current scope	No matching external IOC or exact payload-path telemetry was observed from 13:15-14:00 UTC. The exact path was observed only on FRONTDESK-PC1 in the available telemetry.

Investigation Summary

On 2025-10-15, repeated Windows Security Event ID 4625 failures for Ryan.Adams were observed beginning at approximately 12:52:08 UTC from source workstation DESKTOP-924H12 / 172.16.0.184 against FRONTDESK-PC1.KCD.local. The events showed NTLM authentication via NtLmSsp and were followed by successful Event ID 4624 Type 3 network logons from the same workstation/IP. A broader pivot showed the same source associated with multiple usernames, making the activity consistent with credential guessing and possible password-spray behavior rather than conclusively a single-account brute-force attack.

At 12:57:00 UTC, Sysmon file-creation telemetry showed chrome.exe creating C:\Users\Ryan.Adams\Music\python.exe. At 13:00:33 UTC, Sysmon confirmed execution. One second later, the same process (PID 3080 within the incident window) connected from 172.16.0.110 to 157.245.46.190:8888. The process also resolved ADDC01.KCD.local to 172.16.0.7 and contacted that host on TCP 135 and 49669. These ports are consistent with Windows RPC-related communication, but the available evidence does not demonstrate successful lateral movement or compromise of ADDC01.

Zeek independently confirmed an established TLSv1.3 session from 172.16.0.110 to 157.245.46.190:8888 at approximately 13:03:29 UTC. Suricata also recorded traffic involving the same external IP before python.exe execution; because that telemetry does not provide process attribution, it is retained as a supporting lead rather than attributed directly to python.exe.

At 13:04:59 UTC, PowerShell launched schtasks.exe to create PythonUpdate, configured to execute the suspicious python.exe at system startup under SYSTEM, confirming persistence. At 13:05:53 UTC, Windows Security Center reported Defender entering a SNOOZED state. The timing is security-relevant, but the event does not prove attacker causation.

also does not prove successful lateral movement to ADDC01, does not prove that Defender SNOOZED was attacker-caused, does not identify the exact browser download URL/source, and does not include an independently verified SHA256 for python.exe.